Emmanuel T. Fleurantin

Department of Mathematics • Morehouse College

emmanuel.fleurantin@morehouse.edu • www.emmanuelfleurantin.info

Last Updated: July 2026

AREAS OF INTEREST

Dynamical systems, computational mathematics (numerical methods for boundary value problems, computation of smooth invariant manifolds), computer assisted proofs in dynamics, mathematical biology, data assimilation.

PROFESSIONAL EXPERIENCE

Assistant Professor August 2026-Present
Morehouse College, Atlanta, GA

Postdoctoral Research Fellow January 2022-July 2026
George Mason University, Fairfax, VA

Office of Naval Research (ONR) Postdoctoral Fellow June 2022-December 2022
University of North Carolina at Chapel Hill, Chapel Hill, NC

EDUCATION

PhD in Mathematics May 2018-December 2021
Florida Atlantic University, Boca Raton, FL
- Dissertation title: *Formation, evolution, and breakdown of invariant tori in dissipative systems: from visualization to computer assisted proofs*.

Graduate Certificate in Cyber Security August 2016-May 2018
Florida Atlantic University, Boca Raton, FL

Masters of Science in Mathematics August 2015-May 2018
Florida Atlantic University, Boca Raton, FL
- Thesis title: *On the study of the Aizawa system.*

Masters of Science in Mathematics Education August 2008-August 2010
Nova Southeastern University, Davie, FL

Bachelors of Arts in Economics August 2002-December 2007
University of South Florida, Tampa, FL

PREPRINTS

1. "Predicting the occurrence of Braess paradox in the synchronization threshold of coupled oscillator systems," with Justin Arches, Matt Holzer, Shakeeb Uddin, and Siddhant Sood. Submitted (2026).

2. "Rate-Induced Tipping in a Non-Uniformly Moving Habitat and Determination of the Critical Rate," with Blake Barker, Matt Holzer, Christopher K.R.T. Jones and Sebastian Wieczorek. In revision with SIADS (2026).

3. "Dynamics of Perturbed Elliptic Billiard Tables, with Patrick Bishop, Summer Chenoweth," Jason Mireles-James and Evelyn Sander. In revision with SIADS (2026).

4. "On the Spectrum of Schrödinger Operators Interacting at two Distinct Scales," with Jeremy Marzuola and Christopher K.R.T. Jones. In revision with Journal of Spectral Theory (2026).

PUBLICATIONS

5. “3D Printing of Invariant Manifolds in Dynamical Systems,” with Patrick Bishop, Summer Chenoweth, Alonso Ogueda-Oliva, Evelyn Sander, and Julia Seay. Notices of the AMS, 73 no. 1, doi:10.1090/noti3288 (2026).

6. "Tipping mechanisms in a carbon cycle model," with Katherine Slyman and Christopher K.R.T. Jones. Chaos 35 (2025), no. 5, 053104, doi:10.1063/5.0241550.

7. "Investigating ocean circulation dynamics through data assimilation: a mathematical study using the Stommel box model with rapid oscillatory forcings," with Nathaniel Smith, Anvaya Shiney-Ajay, and Ivo Pasmans. Chaos 34 (2024), no. 10, 103131, doi:10.1063/5.0215236.
8. "A data driven study of the drivers of stratospheric circulation via reduced order modeling and data assimilation," with Julie Sherman, Christian Sampson, Zhimin Wu and Christopher K.R.T. Jones. Meteorology 3 (2024), no. 1, 1–35, doi:10.3390/meteorology3010001.
9. "A dynamical systems approach for most probable escape paths over periodic boundaries," with Katherine Slyman, Blake Barker and Christopher K.R.T. Jones. Physica D 454 (2023), 133860, doi:10.1016/j.physd.2023.133860.
10. "A study of disproportionately affected populations by race/ethnicity during the SARS-CoV-2 pandemic using multi-population SEIR modeling and ensemble data assimilation," with C. Sampson, D. Maes, J. Bennett, T. Fernandes Nunez, S. Marx, and G. Evensen. Found. Data Sci. 3 (2021), no. 3, 479–541, doi:10.3934/fods.2021022.
11. "Computer assisted proofs of two-dimensional attracting invariant tori for ODEs," with Maciej J. Capinski and J.D. Mireles-James. Discrete Contin. Dyn. Syst. 40 (2020), no. 12, 6681–6707, doi:10.3934/dcds.2020162.
12. "Resonant tori, transport barriers, and chaos in a vector field with a Neimark-Sacker bifurcation," with J.D. Mireles-James. Commun. Nonlinear Sci. Numer. Simul. 85 (2020), 105226, doi:10.1016/j.cnsns.2020.105226.
13. "A mathematical model based on IC50 curves to predict tumor responses to drugs," with Catherine I. Berrouet, Jacob Nadulek, Sunil Giri, Katarzyna A. Rejniak, and Necibe Tuncer. FAU Undergrad. Res. J. 7 (2018), 18–32.

SELECTED HONORS, AWARDS AND GRANTS

- National Science Foundation (NSF) Grant DMS - 2137947: "Computing Invariant Manifolds and Assimilating Data in Tipping Problems". This grant funded the MPS-Ascend Postdoctoral fellowship, January 2022 - June 2025.
- Rudy Horne Applied Mathematics Lecture, Conference for African-American Researchers in the Mathematical Sciences (CAARMS), Tufts University, June 2024.
- Mathematically Gifted and Black - Society for Industrial and Applied Mathematics (MGB-SIAM) Early Career Fellow, 2024 - 2026.

TEACHING EXPERIENCE

Morehouse College, Atlanta, GA

- Set Theory (HMTH 255) - Instructor — Fall 2026
- Calculus II (HMTH 162) - Instructor — Fall 2026

George Mason University, Fairfax, VA

- Numerical Analysis I (MATH 446) - Instructor — Summer 2026
- Numerical Analysis II (MATH 447/ CDS 410) - Instructor — Spring 2026
- Mathematical Statistics (MATH 352) - Instructor — Fall 2025
- Computational Dynamics (MATH 689) - Instructor — Spring 2024
- Introduction to Applied Math (MATH 314) - Instructor — Spring 2023
- MEGL Undergrad Research (MATH 491) - Instructor — Fall 2022, Fall 2024, Spring 2025, Fall 2025, Spring 2026, Summer 2026

Florida Atlantic University, Boca Raton, FL

- Introduction to Calculus with Applications (MAC 2210) - Instructor — Fall 2021
- Intermediate Algebra (MAT 1033) - Instructor — Summer 2021
- Methods of Calculus (MAC 2233) - Instructor — Spring 2017, Spring 2019, Spring 2021
- Calculus and Analytic Geometry 1 (MAC 2311) - Instructor — Fall 2020
- Matrix Theory (MAS 2103) - Teaching Assistant — Summer 2020
- Introductory Statistics (STA 2023) - Instructor — Fall 2017, Spring 2018, Spring 2020

- College Algebra (MAC 1105) - Instructor — Fall 2016, Fall 2019
- Calculus and Analytic Geometry 3 (MAC 2313) - Teaching Assistant — Summer 2019
- Engineering Math I (MAP 3305) - Teaching Assistant — Summer 2019
- Precalculus Algebra (MAC 1140) - Instructor — Summer 2018, Fall 2018
- Calculus and Analytic Geometry 1 (MAC 2311) - Teaching Assistant — Fall 2018
- Trigonometry (MAC 1114) - Instructor — Summer 2017

ADVISING AND TRAINING

- **Faculty mentor** of the Mason Experimental Geometry Lab (MEGL) project *Graphons and Network Dynamical Systems,* supervising four undergraduate students during fall 2025, Spring and Summer 2026. The project aimed to use graphons to infer the dynamics of and make rigorous proofs regarding some dynamical systems on large random networks. Students presented their work at George Mason University and the *Geometry Lab United Conference* in June of 2026. A preprint was submitted for publication.
- **Faculty mentor** of the Mason Experimental Geometry Lab (MEGL) project *Existence and Stability of Traveling Waves in Heterogeneous Reaction-Diffusion Systems,* supervising three undergraduate students and one graduate student during fall 2024 and spring 2025. The project studied climate-driven habitat movement effects on population dynamics via reaction-diffusion models. The students presented their work at George Mason University and at the *SIAM Conference on Applications of Dynamical Systems* in May of 2025.
- **Faculty mentor** of a project on using data-assimilation on a low-order climate model where I supervised one undergraduate student and one graduate student during weekly meetings. We conducted this project throughout the fall semester of 2023 (and part of spring 2024) and published our findings in *Chaos: An Interdisciplinary Journal,* in October of 2024.
- **Faculty mentor** of the Mathematics Climate Research Network (MCRN) summer school and research program. Details can be found here: https://aimath.org/workshops/upcoming/mcrn2023/. This was a summer program in July-August of 2023 where I taught intro to dynamical systems and data assimilation mini-courses, provided relevant Python codes and held daily office hours. Finally, for the last week, I lead a research group on a project on ocean circulation.
- **Faculty mentor** of the Mason Experimental Geometry Lab (MEGL) project *Tipping in Climate Impact Models,* where I supervised 2 undergraduate students and 2 graduate students on a project related to tipping in low-dimensional climate models for the fall semester of 2022. The students presented their work both at George Mason University and at the *Overcoming the Computational Complexity of Large Dynamical Systems with Parallel Computations* mini-conference hosted at Georgia Tech in December of 2022.

COLLOQUIA, SPECIAL SESSIONS, MINISYMPOSIA AND WORKSHOP ORGANIZATION

- **Co-Organizer** of the special session on *The Shape of Motion: Bridging Dynamical Systems and Topological Data Analysis* at the Joint Mathematics Meetings (JMM) in January 2027 in Chicago, IL.
- **Organizer** of *Dynamics, Networks, and Complex Systems: A Minisymposium Celebrating Diversity in Mathematics* at the third joint SIAM/CAIMS annual meetings (AN25).
- **Co-Organizer** of the minisymposium on *Tipping in Complex Systems - Theory and Applications* at the 2025 SIAM Conference on Applications of Dynamical Systems.
- **Co-Organizer** of the *Mathematics and Climate Research Network (MCRN) Colloquium Series,* a recurring online forum featuring presentations on mathematical concepts with applications in climate sciences. A list of speakers and abstracts can be found here: https://tinyurl.com/MCRNcolloqium.
- **Co-Organizer** of the minisymposium on *Stability of Traveling Waves - Theoretical and Numerical Methods* at the 2024 SIAM Conference on Nonlinear Waves and Coherent Structures.
- **Co-Organizer** of the *AMS Special Session on Recent Developments in Nonlinear and Computational Dynamics* at the 2024 Spring Eastern Sectional Meeting hosted at Howard University.
- **Co-Organizer** of the American Institute of Mathematics (AIM) workshop on *Computer assisted Proofs for Stability Analysis of Nonlinear Waves* in June 2023. Details can be found here: https://aimath.org/workshops/upcoming/compproofstability/.
- **Co-Organizer** of the *Society for Industrial and Applied Mathematics (SIAM) Special Session on Applications of the Maslov*

Index at the Joint Mathematics Meetings (JMM) in January 2023 in Boston, MA.

- **Co-Organizer** of the *M@TH Hub Workshop: Overcoming the Computational Complexity of Large Dynamical Systems with Parallel Computations* weekly seminar. This was a weekly hybrid talk targeting senior undergraduate students and beginning graduate students in order to introduce them to various math research fields. Previous recorded talks are available at: https://www.youtube.com/channel/UCcBDxJI7F3Tiz2ZsfjYdVOQ.

- **Co-Organizer** of the *FAU SIAM Student Chapter Colloquium Series*. This was a monthly online talk focusing on the use of mathematics and its applications. Previous recorded talks are available at: https://www.youtube.com/channel/UCLLIwl8E06aAfks2dNKve0g.

SELECTED COLLOQUIUM, CONFERENCE, AND SEMINAR PRESENTATIONS

Invited Talks:

- 2026 AMS Fall Southeastern Sectional Meeting - *Homoclinic Tangles and Validated Numerics for Perturbed Elliptical Billiard Maps*, October 2026.

- 2026 SIAM annual meetings (AN26) - *Analytic Continuation and FFT-based Parameterization for Computing Stable and Unstable Manifolds of Hyperbolic Periodic Orbits in Billiard Maps*, July 2026.

- SIAM Conference on Mathematics of Planet Earth (MPE26) - *Rate-Induced Tipping in a Moving Habitat and Determination of the Critical Rate*, July 2026.

- 2026 AMS Spring Eastern Sectional Meeting - *Survival of a Species in a Moving Habitat*, March 2026.

- SIAM Conference on Analysis of Partial Differential Equations (PD25) - *On the Spectrum of Schrödinger Operators Interacting at Two Distinct Scales*, November 2025.

- 18th Annual Symposium on BEER (Biomathematics and Ecology Education and Research) - Multi-Population SEIR Modeling with Data Assimilation: Uncovering COVID-19 Disparities Beyond Aggregate Statistics, November 2025.

- 2025 joint SIAM/CAIMS annual meetings (AN25) - *Geometric Methods for Computing Most Probable Escape Paths in a Carbon Cycle Model*, July 2025.

- SIAM Conference on Applications of Dynamical Systems (DS25) - *Data-Driven Parameter Estimation in a Reduced-Order Quasi-Geostrophic Model of the Polar Vortex using Ensemble Smoothing with Multiple Data Assimilation*, May 2025.

- 13th IMACS International Conference on Nonlinear Evolution Equations and Wave Phenomena: Computation and Theory - *Counting Gap Eigenvalues for the Perturbed 3D Nonlinear Schrödinger Equation by a Maslov Index*, April 2025.

- 2024 SIAM Conference on Nonlinear Waves and Coherent Structures - *Computation of Indices for Stability of Nonlinear Waves*, June 2024.

- 2024 AMS Spring Eastern Sectional Meeting - *Analytical and Computational Techniques for Noise-Induced Transitions over Periodic Boundaries*, April 2024.

- 16th Annual Symposium on BEER (Biomathematics and Ecology Education and Research) - *Parameter Estimation in Epidemiological and Climate Models Using Ensemble Smoothing with Multiple Data Assimilation*, November 2023.

- International Congress on Industrial and Applied Mathematics (ICIAM) - *A Dynamical Systems Approach for Most Probable Escape Paths over Periodic Boundaries*, August 2023.

- 13th AIMS Conference on Dynamical Systems, Differential Equations and Applications - *On the Interplay of Transient Dynamics and Noise-Induced Tipping*, June 2023.

- SIAM Conference on Applications of Dynamical Systems (DS23) - *The Maslow Index and Noise-Induced Tipping*, May 2023.

- Joints Mathematics Meetings 2023 - *The Maslow Index and Noise-Induced Tipping*, January 2023.

- SIAM Conference on Mathematics of Planet Earth (MPE22) - *Using the Maslov Index in Noise-Induced Tipping*, July 2022.

- Dynamics, Topology and Computations 2022 (DyToComp 2022) - *A Dynamical Systems Approach to Most Probable Escape Paths in Non-Gradient Systems*, June 2022.

- AMS Spring Southeastern Sectional Meeting – *Computing Lyapunov Subcenter Manifolds (LSMs) for Hamiltonian Systems*, March 2021.
- CRM CAMP in Nonlinear Analysis – *Computer-assisted proofs of two-dimensional attracting invariant tori for ODEs*, December 2020.

Seminars and Colloquia:

- Mathematics Colloquium, George Mason University - *Spectral Analysis of Schrödinger Operators with Two-Scale Potentials*, October 2025.
- Mathematics and Climate Research Network (MCRN) Colloquium Series - *On the Spectrum of Schrödinger Operators Interacting at Two Distinct Scales*, October 2025.
- Mathematics and Climate Research Network (MCRN) Colloquium Series - *The Maslov Index in Dynamical Systems and PDEs: Recent Applications and Connections to Stability*, September 2024.
- Dynamical Systems Seminar at Imperial College London (DynamIC) - *Investigating Most Probable Escape Paths over Periodic Boundaries: A Dynamical Systems Approach*, June 2024.
- Iowa State University Computational and Applied Math (CAM) Seminar - *On the Study of Noise-Induced Transitions over Periodic Boundaries in Non-Gradient Systems*, April 2024.
- Virginia Commonwealth University Biomath Seminar - *Recent Analytical and Computational Tools for Noise-Induced Transitions over Periodic Boundarie*s, September 2023.
- CDSNS colloquium, Georgia Tech - *A Dynamical Systems Approach for Most Probable Escape Paths over Periodic Boundaries*, March 2023.
- Applied Mathematics Seminar, United States Naval Academy - *A Study of Disproportionately Affected Populations by Race/Ethnicity during the SARS-CoV-2 Pandemic using Multi-Population SEIR Modeling and Ensemble Data Assimilation*, December 2022.
- Applied Mathematics Seminar, George Mason University - *A Dynamical Systems Approach to Most Probable Escape Paths in Non-Gradient Systems*, April 2022.
- Applied Mathematics Seminar, Brigham Young University - *Formation, evolution, and breakdown of invariant tori in dissipative systems: from visualization to computer assisted proofs*, February 2022.
- Graduate Student Seminar at Florida Atlantic University– *Stable/Unstable Manifold Bubbles, Resonant Tori, and Torus-Chaos in the Aizawa System*, January 2019.

Contributed Talks:

- Workshop on the Mathematics of Climate Tipping Points and their Impacts - *Understanding Tipping Phenomena via the Maslov index*, March 2025.
- International Workshop on Non-Autonomous Dynamics in Complex Systems (NADCOM23) - *Recent Analytical and Computational Tools for Noise-Induced Transitions over Periodic Boundarie*s, October 2023.
- SIAM Conference on Applications of Dynamical Systems - *Computing Lyapunov Subcenter Manifolds (LSMs) for Hamiltonian Systems*, May 2021.
- 2nd SIAM Knights Conference – *High order approximation of the center manifold for the Henon-Heiles system*, December 2020.
- Summer School on Dynamics, Data and the COVID 19 Pandemic – *Interpopulation Mixing with Applications in a Two-Population SEIR Model: Using Age Stratification as a Proxy for Racial Disparity in COVID-19 Spread Within a Region (with Nonlinear Ensemble Data Assimilation)*, July 2020.
- 45th Annual New York State Regional Graduate Mathematics Conference – *Resonant Tori in a Vector Field with a Neimark-Sacker Bifurcation*, March 2020.

Poster Presentations:

- *Ensemble Smoother with Multiple Data Assimilation (ESMDA) Applications in Epidemiology and Atmospheric Science*
    - Data Meets Dynamics: Workshop on Data Assimilation for Complex Systems and Applications, August 2025.

    - ICERM Patterns, Dynamics and Data in Complex Systems Workshop, January 2025.

- *The Maslov Index in Dynamical Systems: Recent Applications and Connections to Stability*
    - SIAM Washington-Baltimore Section Fall Meeting, December 2024.
    - ICERM Blackwell-Tapia Conference, November 2024.

- *A Data Driven Study of the Drivers of Stratospheric Circulation via Reduced Order Modeling and Data Assimilation*
    - Mathematics in Digital Twins (MATH-DT Workshop), December 2023.

- *A Dynamical Systems Approach for Most Probable Escape Paths over Periodic Boundaries*
    - Ascend Fellowship Conference, February 2023.

- *Frontline Communities and SARS-CoV-2 - Multi-population Modeling With an Assessment of Disparity by Race/ Ethnicity Using Ensemble Data Assimilation*
    - Second Place Award, Physical Sciences and Mathematics Division - Florida Atlantic University Graduate and Professional Student Association Research Day, April 2021.

- *Transport Barriers, Resonance Tori, and Torus-Chaos in a Vector Field with a Neimark-Sacker Bifurcation*
    - Florida Atlantic University Graduate and Professional Student Association Research Day, April 2019.

PROFESSIONAL SERVICE

- **External Dissertation Committee Member**, Department of Mathematical Sciences, George Mason University (2026-present). Reviewing and evaluating a doctoral dissertation on geometric singular perturbation methods and traveling wave stability.

- **Advisory Board Member**, ASCEND Mentor Network (December 2024-May 2025). Serve on advisory committee for Carnegie Mellon University-led initiative supporting NSF postdoctoral fellows' professional development and mentorship programs.

- **Qualifying Exam Committee Member**, Department of Mathematical Sciences, George Mason University (Summer 2024). Developed and evaluated computational dynamics questions; contributed to overall candidate assessment.

- **President**, Society of Industrial and Applied Mathematics local student chapter (August 2020-August 2021). Led monthly chapter meetings, coordinated research colloquium series, and organized academic events to foster mathematical sciences community engagement.

- **Journal Referee:** Chaos: An Interdisciplinary Journal of Nonlinear Science**,** Communications in Nonlinear Science and Numerical Simulation (CNSNS), International Journal of Bifurcation and Chaos (IJBC), SIAM Journal on Applied Dynamical Systems (SIADS), Physica D: Nonlinear Phenomena, Mathematical Biosciences, Nonlinearity.

PROFESSIONAL AND RESEARCH COMMUNITY MEMBERSHIPS

- American Mathematical Society (AMS) - member since 2016.
- Society for Industrial and Applied Mathematics (SIAM) - member since 2016.
- Mathematics and Climate Research Network (MCRN) - member since 2020.
- Mathematical Association of America (MAA) - member since 2025.

COMPUTER SKILLS

I have experience with:

- Programming Languages: MATLAB (expert), C++ (intermediate), Python (advanced), Mathematica (intermediate), Fortran (intermediate).
- Markup/Typesetting: LaTeX (expert), HTML/CSS (advanced).
- Operating Systems: Linux (advanced).

References available upon request